Digital Humanities and Social Sciences (DHSS) in Israel
21 May 2024
Tel Aviv University

Universal Dependencies for the Queen Puduheba

The project aims to develop a Universal Dependencies (UD) treebank for Hittite, the most ancient Indo-European language attested in cuneiform from the II millenium BC. As a case study, the letter of the Hittite Queen Puduheba (CTH 176, KUB 21.38) was parsed and annotated for UD, containing 1,248 words (193 clauses).
The development of Hittite UD treebanks naturally meets certain challenges that we discuss below, such as annotation of clitic chains, complex sentences, null auxiliaries. This work builds on previous projects [1], [2], [3], [4].

Authors
Emma Yavasan (1), Maria Molina (2)*
emma.yavasan@msmail.ariel.ac.il
mashenke@tauex.tau.ac.il

Affiliations
1 - Ariel University, Digital Pasts Lab
2 - Tel Aviv University, Department of Ancient Near East Cultures and Archaeology

**corresponding author*

Introduction

Universal Dependencies (US) is a framework for consistent annotation of grammar across different human languages. Hittite, however, still lacks a proper UD annotation. For now, there is only one treebank of Hittite built upon some data retrival from the classical grammar of Hittite [5]. This project is an on-going work for developing a consistent treebank of the Hittite letters and instructions.
As a first approach, the Letter of the Queen Puduheba to the Egyptian pharaoh Ramses II has been chosen - as one that is long enough and better preserved than most of the other letters and instructions from the Hittite Kingdom.

The King Hattusili III and his wife Puduheba

Photo: Krähenstein CC BY-SA 3.015

Challenges

There are several challenges in the annotation of the Hittite UD. To mention a few:

- Hittite **correlatives**: a part of the complex sentence or an independent clause?
- Hittite **clitic chains**: independent wordforms or one part together with the host?
- Hittite **lemmatization**: why on Earth did they write in three languages, Hittite, Akkadian and Sumerian?

Methodology

The corpus of texts includes Hittite letters, as published by [6] and represented in [2]. Files are being annotated in CONLL-U format. Visualization of the treebank: PALMYRA 3.0 configurable platform ([7], [8]).

Annotated files and screenshots of generated trees can be downloaded from GitHub: https://github.com/mashenkeisraeli/hittite_ud

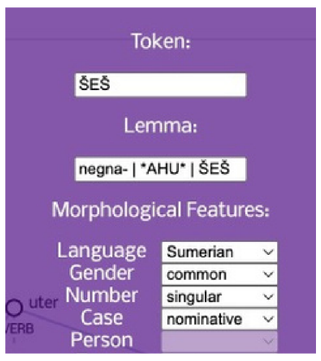
Presentation

Three examples from the "Letter of Puduheba" were chosen to represent the project:

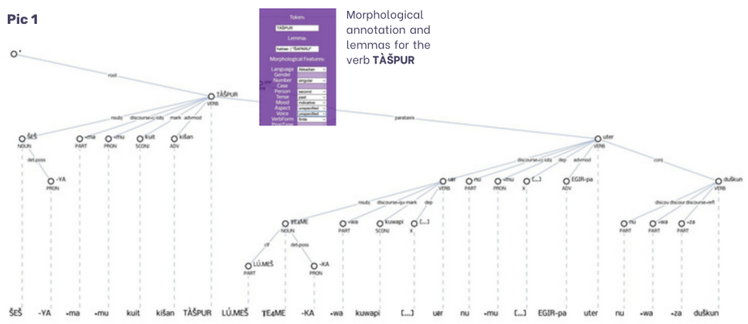
1. ŠEŠ-*YA*=ma=mu kuit kišan *TÀŠPUR* LÚ.MEŠ *ṬE4ME-KA*=wa kuwapi [...] uēr nu=mu [...] EGIR-pa uter nu=wa=za duškun (Pic. 1.)
2. SAG.DU-i *ŠA* ŠEŠ-*YA* TI-tar šuwaru *QATAMMA* ēšdu (Pic. 2.)
3. *ANA* ŠEŠ-*YA* ammel aššulan ammel unuwašḫan uppaḫḫun (Pic. 3.)

Analysis

Speaking of the challenges, Hittite scribes used Akkadian and Sumerian heterograms for Hittite words. That is why we suggested three slots for lemmas in our treebank.



Pic 1

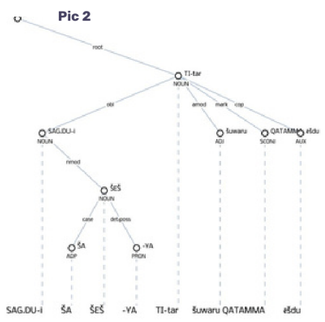


Morphological annotation and lemmas for the verb **TÀŠPUR**

"Concerning the fact that you, my brother, wrote to me as follows: At the time when your messengers came, they brought back to me gifts, and I rejoiced"

This sentence combines four clauses, three of which represent **direct speech** with the first sentence representing the speaker. The problem of annotation for direct speech in Hittite was solved through parataxis with the **mark** (**kuit "as"**) serving *here* as a marker of direct speech.
The **subordinate** clause in the direct speech "When messengers came" includes the adverb **kuwapi "while/when"**, which serves as a **mark** for the main clause: "(They) brought me [...]".
The clitic particle **-wa**, marking **direct speech** in the subordinate clauses, is dependent on the verb in the subordinates and is annotated as **discourse:quot**. The clitic **-ma**, marking a new topic, is annotated as **discourse:conn**. The idea of discourse particles has been suggested in the basic UD guidelines for the relations representing the syntactic links to the wider context. The same applies to the clitic particle **nu**, whose function is basically just to start the sentence inside the paragraph.

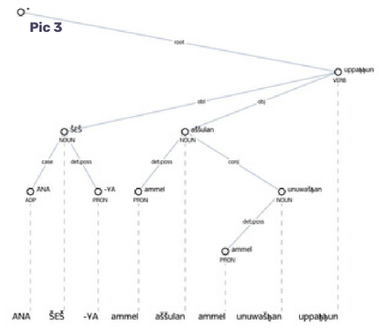
Pic 2



"May the person of my brother likewise enjoy full life!"

For **null auxiliaries**, the UD guidelines suggest that the root is the non-verbal part of the predicate. The copula (**ēš- "to be"**) is dependent on this nominal part and annotated as **"cop"**.
Possessive pronouns ***=mis***, ***ammel***, ***tuel***, as well as their Akkadian counterparts ***-YA, -KA, -SUNU***, etc. receive the **dep:poss** relation to their nouns.
Prepositional phrases (PPs), - keeping in mind that in case of Hittite they are *post*positional, - have nouns as their heads, while prepositions get the **case** relation. This relation is also used with Akkadian prepositions (*ANA*, *INA*, *ŠA*, etc.), though the general understanding is that they were not pronounced, cases being expressed by the form of the noun.

Pic 3



"I have sent my greetings (and) my ornaments to my brother"

When there are two or more **coordinated objects** in the clause, the UD guidelines suggest using only one **obj** relation, with other objects being covered by **obl** or **iobj** depending on the semantics. Alternatively, as shown in Pic. 3, the **conj** relation can also be used to indicate conjunction (without **cc** in this case, with **cc** if there is a conjunction in the sentence).

Morphological annotation for the verb **uter** (see in Pic. 1)

The seal of Hattusili III and Queen Puduheba

Conclusion

For the time being the corpus of Hittite clauses consists of 105 texts, 2,197 clauses, which include approximately 10,000 words. The work is ongoing, with a promising perspective to eventually create the first Hittite treebank based on the language of real documents.

The results of the UD annotation for the "Letter of Queen Puduheba" have allowed for the development of new guidelines for Hittite annotation without requiring many changes to the basic UD guidelines. This will facilitate the easier inclusion of Hittite into the Universal Dependencies system.

References

1. G.Inglese. 2015. Towards a Hittite Treebank. Basic Challenges and Methodological Remarks. *In: Proceedings of the Workshop on Corpus-Based Research in the Humanities (CRH).*
2. M.Molina. 2015-. Annotated Corpus of Hittite Clauses. *http://hittitecorpus.com*
3. G.Inglese, M.Molina, H.Eckhoff. 2018. Incorporating Hittite into PROIEL: a pilot project. *In: Proceedings of the Second Workshop on Corpus-Based Research in the Humanities (CRH-2).*
4. E.Andersen, B.Rozonoyer. 2020. A Small Universal Dependencies Treebank for Hittite. *In: Proceedings of the Fourth Workshop on Universal Dependencies (UDW 2020).*
5. H.Hoffner, H.C.Melchert. 2008. A Grammar of the Hittite Language. Eisenbrauns, Winona Lake.
6. H.Hoffner. 2009. Letters from the Hittite Kingdom. Atlanta: Society of Biblical Literature.
7. T.Javed, N.Habash, D.Taji. Palmyra: A Platform Independent Dependency Annotation Tool for Morphologically Rich Languages. In: The 11th Language Resources and Evaluation Conference (LREC 2018), Miyazaki, Japan. (http://www.lrec-conf.org/proceedings/lrec2018/pdf/333.pdf)
8. D.Taji, N.Habash. Palmyra 2.0: A Configurable Multilingual Platform Independent Tool for Morphology and Syntax Annotation. In: Proceedings of Universal Dependencies Workshop (UDW) 2020 (https://aclanthology.org/2020.udw-1.19.pdf)